

Midterm Project

Augustus Peine

1. Foreign Key Constraint Issues

One of the first challenges occurred when attempting to delete items such as makes, types, and classes. The database prevented deletion due to foreign key constraints because vehicles were still referencing those records.

Solution:

Validation functions were added to check if a make, type, or class was being used by any vehicles before allowing deletion. If so, a user-friendly message was displayed instead of a system error.

2. Filtering Functionality Limitations

Initially, filtering only worked for one category at a time (make OR type OR class). This limited the usability of the application.

Solution:

A more flexible filtering system was implemented using a single function that accepts multiple optional parameters. This allowed users to filter by multiple categories simultaneously.

3. Maintaining Filter and Sort State

When users applied filters or sorting, their selections were lost after submitting forms, leading to a poor user experience.

Solution:

Hidden form inputs were used to preserve the selected filters and sorting options between requests, allowing users to continue refining their search without resetting.

4. Admin Navigation Design

Each admin page needed navigation links that did not include the current page, which required dynamic control over displayed links.

Solution:

Custom navigation sections were created for each admin page, excluding links to the current page to improve usability and navigation clarity.

Conclusion

This project provided valuable experience in building a full PHP application using MVC principles. The challenges encountered helped strengthen understanding of database relationships, user interface design, and structured programming.